

## Tutorial 4: Second-Order Linear ODEs (Part I)

[Solution]

### A) Linear Dependence/Independence & Wronskian

1. Determine if the set of functions is linearly independent or dependent.

$$\text{a) } \{\sin x, \cos x\}, \quad \text{b) } \{1, x, x^2\}, \quad \text{c) } \{1 - x, 1 + x, 1 - 3x\}$$

2. Calculate  $W(x^3, |x^3|)$  on  $[-1, 1]$ . Is  $\{x^3, |x^3|\}$  linearly dependent on  $[-1, 1]$ ?

Can we say if the Wronskian of two functions is zero, then they are LD?

**Solution:** a) The ratio is not a constant, so they are LI.

b) We can use the definition or determine by using the Wronskian. To use the definition, we form the equation

$$c_1 + c_2x + c_3x^2 = 0,$$

and find readily that it holds for all  $x \in (-\infty, \infty)$ , if and only if  $c_1 = c_2 = c_3 = 0$ , so they are LI. On the other hand, we can calculate its Wronskian

$$W(1, x, x^2) = \det \begin{pmatrix} 1 & x & x^2 \\ 0 & 1 & 2x \\ 0 & 0 & 2 \end{pmatrix} = 2,$$

so they are LI.

c) We form the equation

$$c_1(1 - x) + c_2(1 + x) + c_3(1 - 3x) = 0,$$

i.e.,

$$(c_1 + c_2 + c_3) + (-c_1 + c_2 - 3c_3)x = 0,$$

So we have

$$c_1 + c_2 + c_3 = 0, \quad -c_1 + c_2 - 3c_3 = 0,$$

which has non-trivial solutions. Hence,  $\{1 - x, 1 + x, 1 - 3x\}$  is LD.

2. If  $0 \leq x \leq 1$ , we have

$$W(x^3, |x^3|) = W(x^3, x^3) = 0,$$

while if  $-1 \leq x < 0$ ,

$$W(x^3, |x^3|) = W(x^3, -x^3) = 0.$$

We can use definition to show that  $x^3$  and  $|x^3|$  are LI. The statement: “ $W(f, g) = 0 \Rightarrow f, g$  are LD” is False.

## B) Wronskian & Abel’s Formula

Let  $y_1$  and  $y_2$  be two solutions to the equation

$$y'' + p(x)y' + q(x)y = 0, \quad x > 0, \tag{1}$$

where  $p(x)$  and  $q(x)$  are continuous for all  $x > 0$ . Suppose that  $y_1 = \frac{\sin x}{\sqrt{x}}$ , and the

Wronskian of  $y_1$  and  $y_2$  is  $W(x) = \frac{1}{x}$ .

(i) Derive the Abel’s equation,

$$W'(x) + p(x)W(x) = 0$$

and obtain the Abel’s formula of  $W$  in terms of  $p$ .

(ii) Find the functions  $p(x)$  and  $q(x)$  in the equation (1).

(iii) Find the general solution of the equation (1).

**Solution:** (i). Let  $W(x) = W(y_1, y_2)(x)$  be the Wronskian of two solutions  $y_1, y_2$ . Note that

$$W(x) = \det \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} = y_1 y_2' - y_1' y_2,$$

so we have

$$\begin{aligned} W'(x) &= (y_1 y_2' - y_1' y_2)' = (y_1 y_2'' + y_1' y_2') - (y_1'' y_2 + y_1' y_2') \\ &= y_1 y_2'' - y_1'' y_2 = y_1(-p y_2' - q y_2) - (-p y_1' - q y_1) y_2 \\ &= -p y_1 y_2' + p y_1' y_2 = -p(y_1 y_2' - y_1' y_2) = -p(x)W(x). \end{aligned}$$

Then

$$W'(x) + p(x)W(x) = 0.$$

(ii) From the Abel's formula, we have

$$p(x) = -\frac{W'(x)}{W(x)} = \frac{1}{x}.$$

Since  $y_1$  is a solution, we have

$$y_1'' + p y_1' + q y_1 = 0 \Rightarrow q = -\frac{y_1'' + p y_1'}{y_1} = 1 - \frac{1}{4}x^{-2}.$$

Therefore, the DE (1) is

$$y'' + \frac{1}{x}y' + \left(1 - \frac{1}{4}x^{-2}\right)y = 0, \quad x > 0.$$

(iii) We next look for  $y_2$  and note

$$W(x) = y_1 y_2' - y_1' y_2 = \frac{1}{x}.$$

This implies

$$y_2' - \frac{y_1'}{y_1} y_2 = \frac{1}{x y_1},$$

that is

$$y_2' + \left(\frac{1}{2x} - \cot x\right)y_2 = \frac{1}{\sqrt{x} \sin x}.$$

Then we solve the first-order linear DE. Indeed, the IF

$$\begin{aligned} I(x) &= \exp\left(\int \left(\frac{1}{2x} - \cot x\right) dx\right) = \exp\left(\frac{1}{2} \ln x - \ln |\sin x|\right) \\ &= \frac{\sqrt{x}}{|\sin x|} = \pm \frac{\sqrt{x}}{\sin x}. \end{aligned}$$

Here, we take  $I(x) = \frac{\sqrt{x}}{\sin x}$ . Then

$$\begin{aligned} y_2(x) &= \frac{1}{I(x)} \left( \int I(x) \frac{1}{\sqrt{x} \sin x} dx + C \right) \\ &= \frac{\sin x}{\sqrt{x}} \left( \int \frac{1}{\sin^2 x} dx + C \right) \\ &= \frac{\sin x}{\sqrt{x}} (-\cot x + C) = -\frac{\cos x}{\sqrt{x}} + C y_1. \end{aligned}$$

Thus the general solution is

$$y(x) = C_1 \frac{\sin x}{\sqrt{x}} + C_2 \frac{\cos x}{\sqrt{x}}.$$

### C) Reduction of Order

Consider the second-order linear homogeneous equation

$$y'' + p(x)y' + q(x)y = 0,$$

where  $p(x)$  and  $q(x)$  are continuous functions on some interval  $I$ .

- (a) Show that if there exists a constant  $r$  such that  $r^2 + rp(x) + q(x) = 0$  for all  $x \in I$ , then  $y(x) = e^{rx}$  is a solution to the above equation.
- (b) Solve this second-order equation with

$$p(x) = -2\left(1 + \frac{1}{x}\right), \quad q(x) = 1 + \frac{2}{x}.$$

**Solution:** (a) Notice that  $e^{rx}$  is a solution, when

$$\begin{aligned} (e^{rx})'' + p(x)(e^{rx})' + q(x)e^{rx} &= (r^2 + rp(x) + q(x))e^{rx} = 0 \\ \iff r^2 + rp(x) + q(x) &= 0 \text{ for all } x \in I. \end{aligned}$$

(b) For this equation, we find that

$$\begin{aligned} r^2 + r \left( -2 \left( 1 + \frac{1}{x} \right) \right) + 1 + \frac{2}{x} &= 0 \\ \iff (r^2 - 2r + 1) + \frac{2}{x}(1 - r) &= 0 \end{aligned}$$

Therefore, we have  $r = 1$ , so  $y_1 = e^x$  is a solution of this equation. Then we can use the reduction of order to find  $y_2$  through the formula

$$y_2 = y_1 \int \frac{1}{y_1^2} e^{-\int p(x) dx} dx = e^x \int e^{-2x} e^{2\int(1+\frac{1}{x})dx} dx = e^x \int x^2 dx = \frac{1}{3} x^3 e^x.$$

The GS is

$$y(x) = (C_1 + C_2 x^3) e^x.$$

#### D) Second-Order Linear Homogeneous DE

1. Verify that  $y_1 = x^2$  is a solution to  $x^2 y'' - 3xy' + 4y = 0$ . Find the general solution to this DE.

**Solution:** 1. We can verify readily that  $y_1 = x^2$  is a solution. Then we rewrite the equation in the standard form

$$y'' - \frac{3}{x} y' + \frac{4}{x^2} y = 0$$

Then using the formula, we find

$$y_2 = y_1 \int \frac{1}{y_1^2} e^{-\int p(x) dx} dx = x^2 \int \frac{1}{x^4} e^{3\int \frac{1}{x} dx} dx = x^2 \ln x.$$

So the GS is

$$y(x) = (C_1 + C_2 \ln x) x^2.$$

2. Solve the following second-order equations

- (i)  $y'' - y' - 2y = 0$ .
- (ii)  $y'' + 6y' + 25y = 0$ .
- (iii)  $y'' + 4y' + 4y = 0$ ,  $y(0) = 1$ ,  $y'(0) = 4$ .

**Solution:**

- (i)  $y(x) = c_1 e^{2x} + c_2 e^{-x}$
- (ii)  $y(x) = e^{-3x} (c_1 \cos 4x + c_2 \sin 4x)$
- (iii)  $y(x) = e^{-2x} + 6x e^{-2x}$

3. Suppose that a second-order linear homogeneous DE with constant coefficients has one solution of the form  $y_1(x) = e^{2x} \sin x$ . Find the general solution to this underlying DE.

**Solution:**  $y(x) = c_1 e^{2x} \sin x + c_2 e^{2x} \cos x$

4. Suppose that a second-order linear homogeneous DE with constant coefficients has one solution of the form  $y_1(x) = x e^{-x}$ . Find the general solution to this underlying DE.

**Solution:**  $y(x) = c_1 x e^{-x} + c_2 e^{-x}$

5. Consider the equation

$$y''(t) + ty'(t) + e^{-t^2} y(t) = 0.$$

Use the change of variable  $x = \int e^{-t^2/2} dt$  to convert the equation to equation with constant coefficients, and find the general solution.

**Solution:** Let  $z(x) = y(t)$ . Then

$$y' = \frac{dy}{dt} = \frac{dz}{dx} \frac{dx}{dt} = e^{-t^2/2} \frac{dz}{dx}$$

and

$$\begin{aligned} y'' &= \frac{d}{dt} \left( e^{-t^2/2} \frac{dz}{dx} \right) = -te^{-t^2/2} \frac{dz}{dx} + e^{-t^2/2} \frac{d}{dt} \left( \frac{dz}{dx} \right) \\ &= -te^{-t^2/2} \frac{dz}{dx} + e^{-t^2/2} \frac{d}{dx} \left( \frac{dz}{dx} \right) \frac{dx}{dt} \\ &= -te^{-t^2/2} \frac{dz}{dx} + e^{-t^2/2} \left( \frac{d^2 z}{dx^2} \right) e^{-t^2/2} \\ &= -te^{-t^2/2} \frac{dz}{dx} + e^{-t^2} \left( \frac{d^2 z}{dx^2} \right) \end{aligned}$$

Then the equation becomes

$$\begin{aligned} &y''(t) + ty'(t) + e^{-t^2} y(t) \\ &= -te^{-t^2/2} \frac{dz}{dx} + e^{-t^2} \left( \frac{d^2 z}{dx^2} \right) + te^{-t^2/2} \frac{dz}{dx} + e^{-t^2} z \\ &= e^{-t^2} \left( \frac{d^2 z}{dx^2} \right) + e^{-t^2} z = 0, \end{aligned}$$

i.e.,

$$\frac{d^2 z}{dx^2} + z = 0 \quad \Rightarrow \quad z = c_1 \cos x + c_2 \sin x$$

Finally, the GS of the original equation is

$$y(t) = c_1 \cos \left( \int e^{-t^2/2} dt \right) + c_2 \sin \left( \int e^{-t^2/2} dt \right).$$

6. Find a suitable transform  $x = v(t)$  to convert the problem

$$ty''(t) + (t^2 - 1)y'(t) + t^3 y(t) = 0,$$

to a DE with constant coefficients and solve it.

**Solution:** Let  $z(x) = y(t)$ . Then

$$y' = \frac{dy}{dt} = \frac{dz}{dx} \frac{dx}{dt} = \frac{dz}{dx} \frac{dv}{dt}$$

and

$$\begin{aligned} y'' &= \frac{d}{dt} \left( \frac{dv}{dt} \frac{dz}{dx} \right) = \frac{d^2 v}{dt^2} \frac{dz}{dx} + \frac{dv}{dt} \frac{d}{dt} \left( \frac{dz}{dx} \right) \\ &= \frac{d^2 v}{dt^2} \frac{dz}{dx} + \frac{dv}{dt} \frac{d}{dx} \left( \frac{dz}{dx} \right) \frac{dx}{dt} \\ &= \frac{d^2 v}{dt^2} \frac{dz}{dx} + \left( \frac{dv}{dt} \right)^2 \frac{d^2 z}{dx^2}. \end{aligned}$$

Hence, we convert the DE into

$$\begin{aligned} ty''(t) + (t^2 - 1)y'(t) + t^3 y(t) &= t \left\{ \frac{d^2 v}{dt^2} \frac{dz}{dx} + \left( \frac{dv}{dt} \right)^2 \frac{d^2 z}{dx^2} \right\} \\ &+ (t^2 - 1) \frac{dv}{dt} \frac{dz}{dx} + t^3 z = 0. \end{aligned}$$

We group the coefficients in terms of the derivatives of  $z$  :

$$t \left( \frac{dv}{dt} \right)^2 \frac{d^2 z}{dx^2} + \left\{ t \frac{d^2 v}{dt^2} + (t^2 - 1) \frac{dv}{dt} \right\} \frac{dz}{dx} + t^3 z = 0.$$

To be a constant coefficient problem, we require and set

$$t \left( \frac{dv}{dt} \right)^2 = t^3 \quad \Rightarrow \quad \left( \frac{dv}{dt} \right)^2 = t^2 \quad \Rightarrow \quad \frac{dv}{dt} = t \quad \Rightarrow \quad v = \frac{t^2}{2}.$$

Now, we substitute it into the coefficient of the first derivative:

$$t \frac{d^2 v}{dt^2} + (t^2 - 1) \frac{dv}{dt} = t^3.$$

In summary, if we choose

$$x = v(t) = \frac{t^2}{2},$$

we can convert the equation to a constant coefficient ODE:

$$z'' + z' + z = 0.$$

Its GS is

$$z(x) = e^{-x/2} \left( c_1 \cos \frac{\sqrt{3}x}{2} + c_2 \sin \frac{\sqrt{3}x}{2} \right),$$

so the GS of the original equation is

$$y(t) = e^{-t^2/4} \left( c_1 \cos \frac{\sqrt{3}t^2}{4} + c_2 \sin \frac{\sqrt{3}t^2}{4} \right).$$